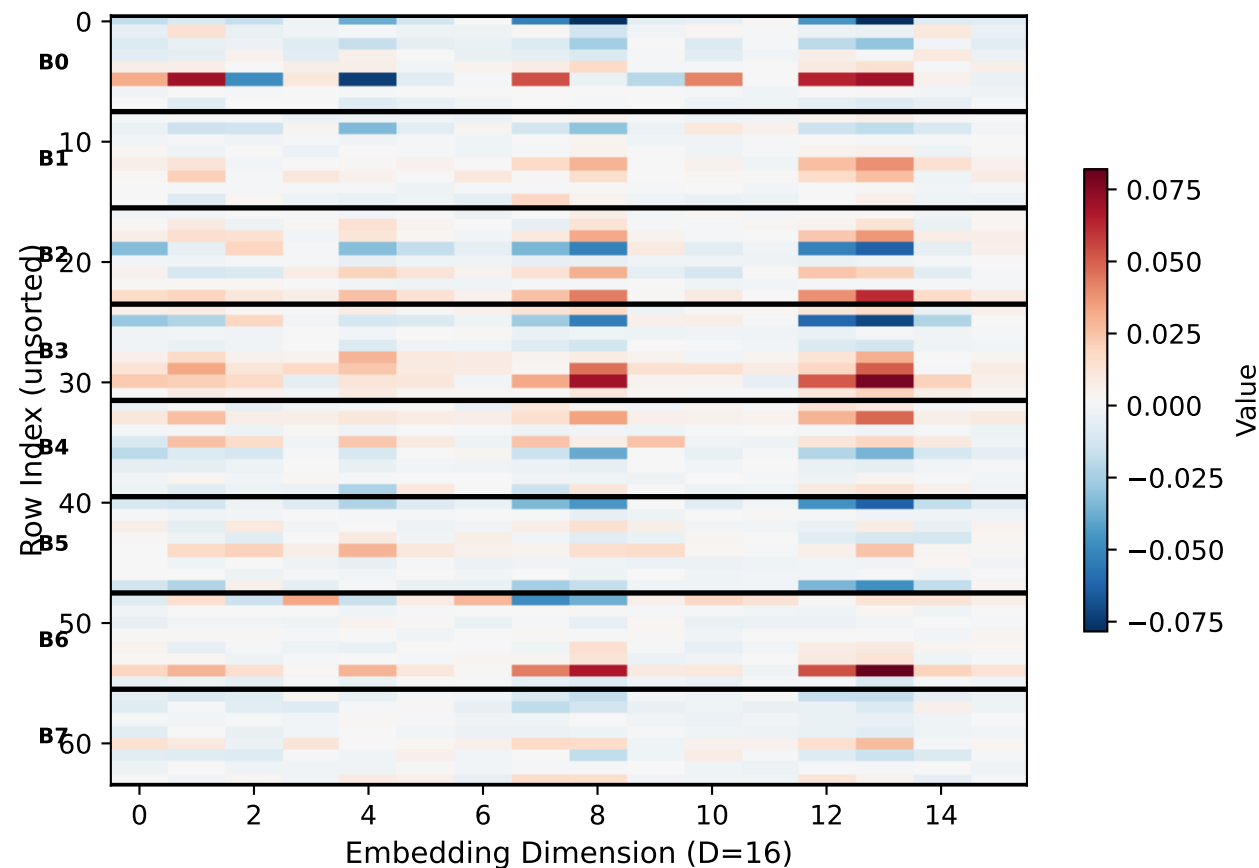


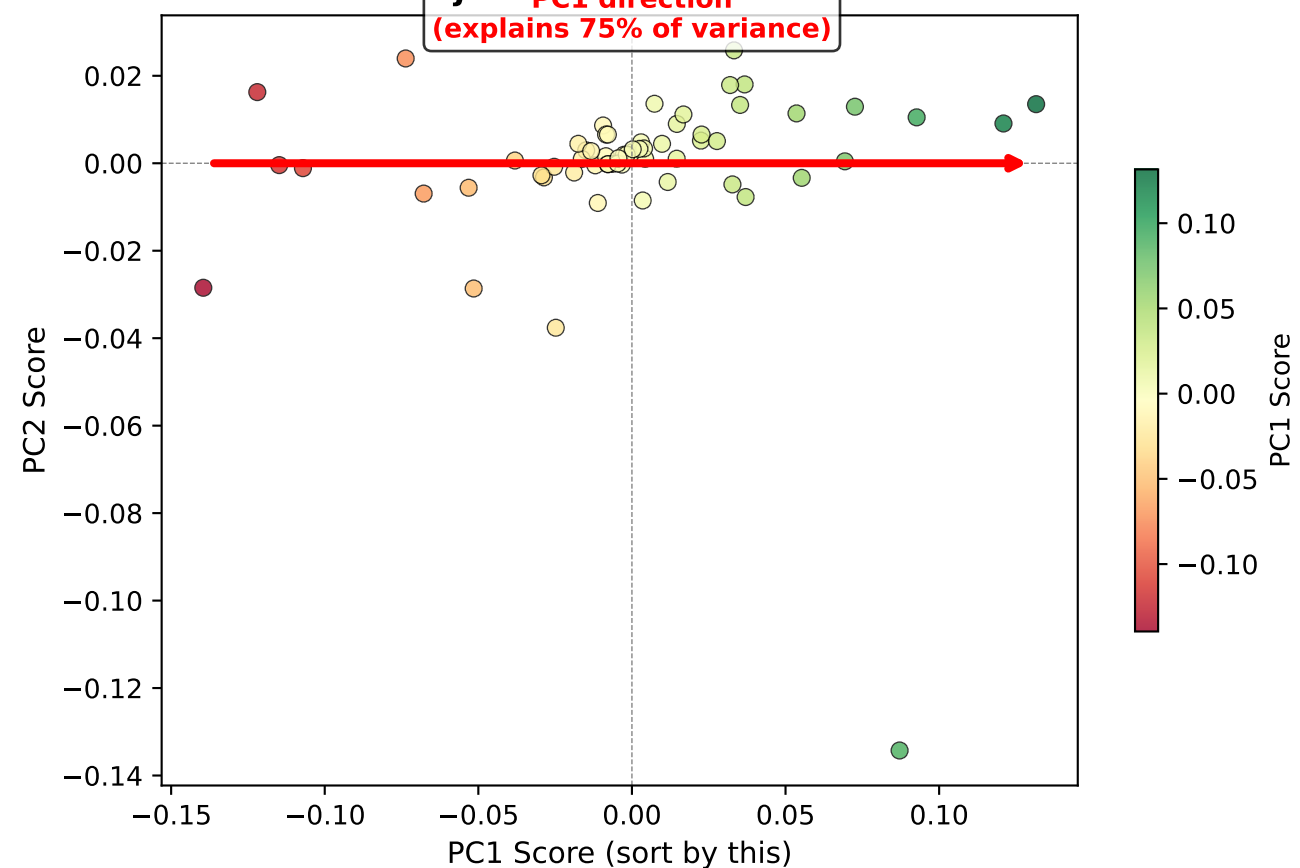
PCA Sort: How It Works

Sort embedding rows by first principal component before computing block means
→ adjacent rows become similar → block mean is a better approximation → less AUC loss

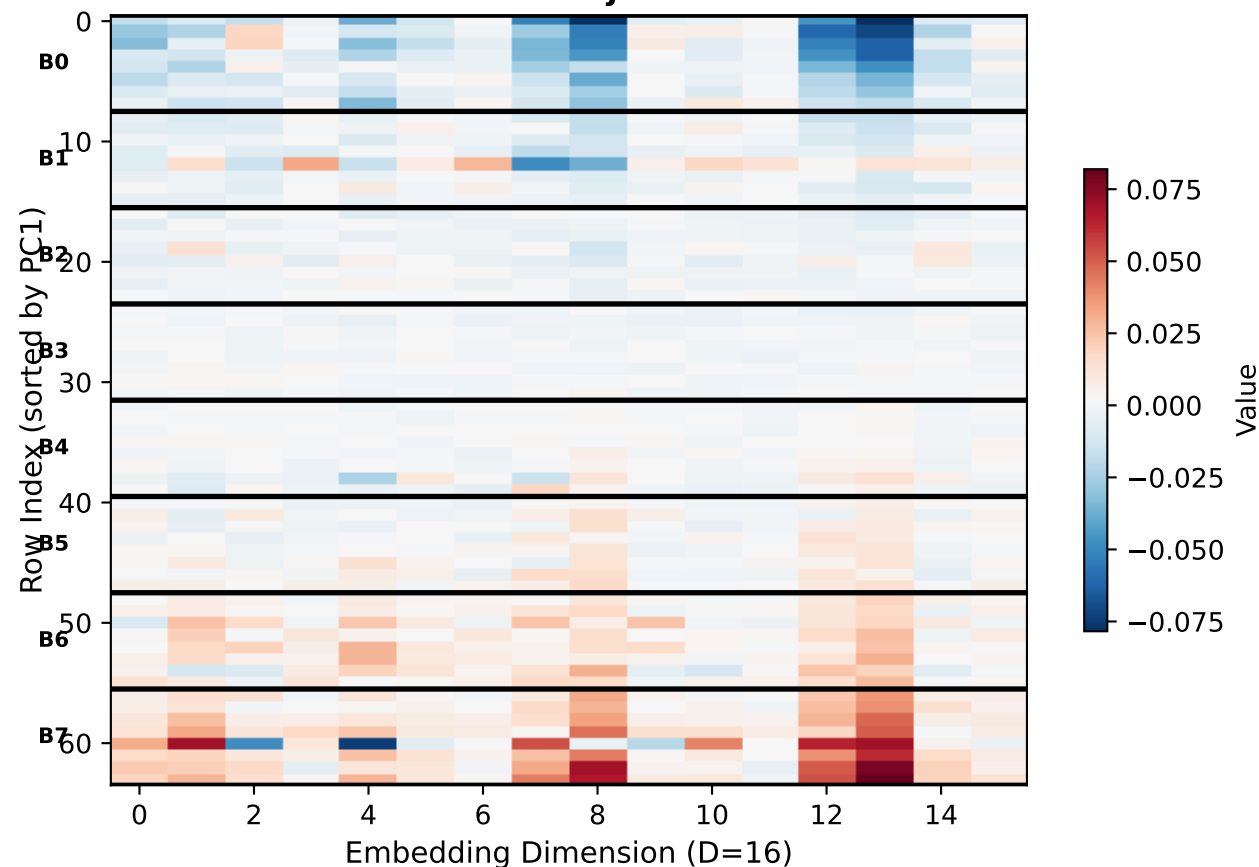
Step 1: Cold Rows (Original Order)
Rows within each block are dissimilar



Step 2: PCA Finds Best Direction
Project all rows onto PC1
PC1 direction (explains 75% of variance)



Step 3: Sort Rows by PC1 Score
Similar rows are now adjacent in each block



Step 4: Block-Mean Error Comparison
PCA sort: 68% less error than unsorted, 9% less than value sort

